Truth-resistant AI

Autoclave bags for truth-resistant AI

An autoclave is a pressure chamber that sterilizes with steam, typically at 121 °C for at least 15 minutes. Wet labs put their waste in red autoclave bags, and the bags go into the autoclave before they leave the building. The autoclave does not care whether a bacterium is resistant to antibiotics. An antibiotic attacks one mechanism of the cell, and a bacterium can acquire a gene that protects that mechanism; this is what the plasmid technique in the Introduction exploits. Steam under pressure attacks every mechanism at once, so the resistant bacterium dies with the others.

In wet labs such as Jeffrey Ishizuka's, physicians take continuing medical education in biohazards and in AI hazards, and an internal joke carries the vocabulary from one to the other: the tools that detect AI hallucinations are called *autoclave bags*. We take the joke seriously, because the analogy goes further than the bag. Some AI errors resist correction the way some bacteria resist antibiotics. Ask the AI "are you sure?" and it apologizes and repeats the error; give it a better prompt and the error returns in other words; ask a second AI to review the first, and it may approve the error. We call an AI in this state *truth-resistant*. Remedies that act through the language model are its antibiotics, and the model can resist them.

The autoclave for truth-resistant AI is a test against reality, which does not negotiate. PERCEPT is such an autoclave [2,6]. When C2S-Scale proposes a hypothesis about how tumor and immune cells respond to a drug [3], Curtis Perry treats the patient's own cells in the wet lab and measures what happens. A hypothesis that fails the measurement is dead, however fluent the model that proposed it. On a smaller scale, running a program kills a wrong claim about what the program does, and a grammar kills a sentence that cannot be parsed.

The two devices have different jobs:

- an **autoclave bag** detects and contains: it marks the suspect output and keeps it from leaving the editor before someone deals with it. RASP's rejection of a sentence, and the red bag in Browsered, are autoclave bags;
- an **autoclave** destroys: it settles whether the output is true by testing it against something outside the model, such as cells, a running program, or a grammar.

A bag without an autoclave only postpones the problem, and an autoclave needs bags, because nobody carries loose waste to it. The human in the loop is the person who closes the bag and runs the cycle.

The analogy also marks its own limits. Prions, the agents of mad cow disease, survive a standard autoclave cycle and need a longer, hotter one. Likewise, some AI errors survive our tests: a wrong sentence that happens to be grammatical

passes RASP (see "Errors we can fix and errors we cannot"). Hence the warning Browsered shows: *Flexner's red AI-hazard bag is a proof of concept. It catches some AI errors and misses others.*

The warning states what the bag does and tells the user nothing to do. A first draft added "Keep using the safeguards your AI vendor recommends", and we removed it for two reasons. First, the safety recommendations vendors post are disclaimers, and a disclaimer meant to assign responsibility is a contract (see "Unsolicited disclaimers as a design problem"); a sentence pointing to them passes the reader to someone else's contract. Second, those recommendations reduce to "check the answer", which the user who most needs an AI cannot do. A general practitioner asks an AI because he does not know the answer; if he could check it, he would not have asked. The check must be done by something he can run, such as an autoclave, and not by knowledge he lacks.

We found no earlier use of the terms *truth-resistant AI* or *autoclave bag* for hallucination detectors. Biosafety containment has been proposed as a model for securing autonomous AI agents [28], with biosafety levels mapped to sandboxing and access control; that work addresses what an agent may do. Our analogy addresses whether what an AI says is true.

Biohazard and AI-hazard

Doctors like Curtis Perry, Jeffrey Ishizuka, Jonathan C. Trent, and Philippos Costa use something called targeted therapy. However, the targets that their drugs aim at are not sitting ducks. Cancer becomes resistant to the drug. We should compare the listed doctors to homing missiles tracking their moving targets.

If traditional targeted therapy acts like a missile fired at a stationary bunker, modern oncology requires 'moving-target therapy'—a system where the therapeutic missile continuously recalculates its trajectory to track a mutating, evasive cell.

"A doctor shouldn't use AI because they hallucinate." This is a naive and stupid statement for two reasons. First, whether we like it or not, doctors will leverage AI. Besides, in the right setup, AI with a human in the loop doesn't hallucinate. Consider the protocol: "C2S-Scale generates testable biological hypotheses, which Curtis Perry then verifies in the wet-lab PERCEPT platform." To produce mutation sequences, Curtis Perry and Philippos Costa introduce the mutation, conjugated with antibiotic resistance, into plasmids. A small percentage of the bacteria acquire the plasmids. After the culture grows, the doctors apply an antibiotic, which kills the bacteria that didn't receive the plasmids. This clever trick gives the doctors a culture where all bacteria produce the mutation. But doctors also receive continuing medical education on biohazards, where they learn protocols to prevent a lab leak into the wild. They also have continuing medical education on Medical AI Safety,

where they learn protocols to prevent AI hallucination from escaping into the wild.

Then, let's do it like Curtis Perry: stop complaining about hallucinations and biohazards and learn how to work with wet-lab and AI while avoiding leaks of biohazards, such as antibiotic resistance and AI hazards, such as hallucinations. To prevent biohazard leaks, we use primary containment devices. To prevent AI hazard leaks, we will use a human-in-the-loop approach.

Here's why we need special editors. For domestic waste, a garbage can will do. Still, for Jeffrey Ishizuka's Lab, we will need a puncture-resistant biohazard sharps container or a designated foot-pedal biohazard bin lined with a red autoclave bag, both prominently marked with the universal biohazard label. In an apartment building, we will need an automatic disposal method, such as a sink grinder. But Jeffrey Ishizuka needs a human in the loop to dispose of biohazards. The human in the loop also needs devices like an autoclave and red autoclave bags. In the case of a human-in-the-loop for disposing of AI-hazards, there are many devices, such as:

1. A special Editor that allows the human in the loop to handle AI hazards. This editor must be extensive and use a scripting language that doesn't change over time, so people worldwide can contribute major modes to protect against computer-induced mistakes. Such an editor exists: Emacs. The scripting language, elisp, has not changed in 40+ years. If I load an AI-hazard disposer written in elisp by Joshua Lederberg in the 1990s, it will work out of the box.
2. GOFAI tools, such as RASP. Thousands of GOFAI tools exist worldwide, and according to Gary Marcus, they may effectively protect against AI-hazards. Usually, you can't use them out of the box, like an elisp major mode. But a Generative AI, such as Claude Code, can retrofit RASP (Robust Accurate Statistical Parser) in less than one minute.

GOFAI tools deserve a section of their own. This Manifest guides my AI into the editor with AI-hazard containment. Although Emacs is the industry standard for AI-hazard containment, Superhuman doesn't support it. That is strange, since the Ukrainian AI tech giant uses both GOFAI tools and Emacs heavily. Then, the Abraham-Flexner group created two editors to replace Emacs. But don't worry. Both editors work exactly like Emacs, including the same crazy keybindings and kill ring. Emacs runs the same way on Windows, macOS, Linux, or a supercomputer. For example, copying or renaming a file, entering a folder and deleting a file require different operations on macOS and Windows. But for Emacs users, the operations are the same on any computer. Our editor, Browsered, tries to preserve this feature.

References

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